The Development of the Application for Public Health Support for the Elderly

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***Abstract*—This research aims to develop a public health care support application for the elderly. The sampling is a group of 30 elderly in Khlong Lad Phrao community, Bangkok by purposive sampling and a group of 15 public health care volunteers and 3 information technology experts. The instruments used in this research are questionnaires and interviews. The development tools of the applications were as follows 1. ASP.NET 2. PHP Apache 3. Java 4. SQL Server. The data analysis statistics are frequency, percentage mean, and standard deviation. The result shows that from the trial of the application for public health support for the elderly, the performance and satisfaction assessment form shows overall assessment results of a mean of 4.57 and a standard deviation of 0.54, implying that the application is the most satisfied. This developed application can be downloaded on Android apps on Google Play named "PHS Application".**

Keywords—elderly, application, public health support, TAM model.

I. INTRODUCTION

Thailand has entered a completely elderly society with more than 13 million people, making it the second country in ASEAN after Singapore. Entering an aging society is an important dynamic that has far-reaching consequences for both global society and Thailand. Particularly, Thailand has entered an aging society since 2005 and is anticipated to transition into an aged society within the next 6 years, or by 2024. The Kasikorn Research Center (2018) estimates that Thailand's elderly population aged 65 and over will increase to 14.0% by 2024 [1] .

Furthermore, estimates indicate that 1 in 10 Thais are 60 years of age or older. Within the next 20 years, projections indicate that this elderly population will rise to 1 in 5 and transform into a "super-aged society". Estimates indicate that by 2040, Thailand's elderly population will rise to 30% of the total population, representing 20 million individuals, or 1 in 3 of the Thai population. Moreover, projections indicate that the number of individuals aged 80 and over could reach up to 3,500,000. The province with the highest elderly population is Bangkok, with over 1 million people (17.98%), followed by Nakhon Ratchasima, Chiang Mai, Khon Kaen, and Ubon Ratchathani [2].

Khlong Lad Phrao is another community located along the Lad Phrao Canal, which is a 24 kilometer long canal consisting of 6 areas (Sai Mai, Don Mueang, Chatuchak, Lak Si, Bang Khen, and Lad Phrao). It is a densely populated community with 4,847.27 people per square kilometer, comprising 74,692 males, 84,822 females, and a total of 159,514 people living in 101,602 houses [3].

Additionally, there is a growing trend in the elderly population, estimated to account for 15% of the total population in the country [4]. Therefore, the government has placed greater importance on the elderly, in line with the vision of the National 20-Year Strategy (2015 - 2034). This includes providing suitable employment, creating an environment and innovations that focus on human resource development, ensuring quality education and learning that align with 21st century learning, and preventing and controlling social determinants of health to create a healthy and appropriate state for everyone. This includes Village Health Volunteer known as "Bangkok Public Health Volunteer", who represent the community.

However, field surveys and interviews with public health volunteer revealed that they continue to encounter difficulties in promptly reaching ill elderly individuals, leading to a delay in basic healthcare and a lack of long-term continuous care for the elderly. Furthermore, the community lacks basic health data and communication tools to coordinate with nearby public and private hospitals in emergencies, which leaves public health volunteer without appropriate and community-aligned resources. Therefore, researchers have recognized the importance of studying the development of an application to support public health for the elderly. This research article will review and analyze the literature, including the Technology Acceptance Model (TAM), to provide insights into the development and implementation of public health support applications for the elderly. This will benefit public health volunteer, reduce the risk of elderly mortality, and enhance the country's public health work in the future.

II. LITERATURE REVIEW

A. Technology Acceptance Model (TAM)

The Technology The Technology Acceptance Model (TAM) explains the factors that influence the intention to use technology. It includes four variables: external variables, perceived usefulness (PU), perceived ease of use (PEOU), and attitude toward using. According to the TAM theory, the

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attitude towards usage is influenced by the perception of the benefits gained from technology and the perception that the system is easy to use. The intention to use behavior is influenced by the attitude towards usage and the perception of the benefits gained from technology, ultimately leading to the acceptance of actual usage. However, to fully understand individual technology acceptance, it is necessary to add variables. Previous research by I.F.R. Green (2005) and V. Venkatesh, F. Davis, (2000) have demonstrated the need to explain people's reasons for perceiving the benefits of information systems.

Technology Acceptance Model (TAM) is a study of human behavior that seeks to explain how and why individuals accept new technology. Davis (1989) developed the model, which evolved from Ajzen and Fishbein's (1980) [5]. Theory of Reasoned Action (TRA). Nevertheless, the Technology Acceptance Model (TAM) is more widely known than the theory of deliberative reasoning. The theory of deliberative reasoning focuses on the general behavior of consumers, while TAM focuses on the attitudes of information technology users (Mathieson et al. 2001). Figure 1 depicts the model structure.

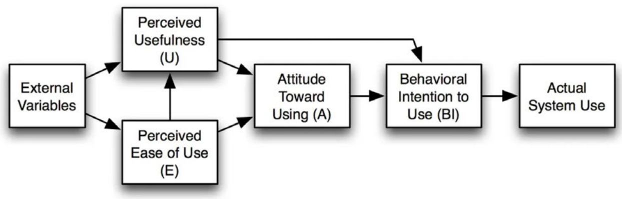

Fig. 1. Technology Acceptance Model (TAM).

The Technology Acceptance Model (TAM) posits that external variables, which shape individual perceptions, influence an individual's acceptance of technology. These variables include knowledge, beliefs, experiences, and social behavior, and they lead to 5 stages in the technology acceptance process.

1) Perceived Usefulness (PU): Refers to the extent to which an individual believes that using a particular technology would enhance their performance.

2) Perceived Ease of Use (PEOU): Refers to the degree to which a person believes that using a particular system would be free from effort.

3) Attitude: Towards using something encompasses behavior, intention to use, and actual use. Allport and G. W. (1968) [6] Attitudes are influenced by a system's perceived usefulness and ease of use. If users perceive a technology as useful and simple to use, they will have a positive attitude towards it. This positive attitude will also influence their intention to use the technology. When users intend to use the technology, they will feel that they should use it the real technology [7].

4) Behavior Intention to Use (BI): Refers to the behavior or interest of users who attempt to use technology and information technology.

5) The Actual System Use: Refers to the process of implementing and embracing technology. This occurs when perceived benefits and ease of use generate interest and intention to use the technology, ultimately leading to its acceptance and actual use.

The literature review leads us to the conclusion that public health volunteers will consistently utilize technology to support the elderly population. Public health volunteers need to have a positive attitude toward this technology. Ajzen & Fishbein (1980) found that attitude influences actions in their Theory of Reasoned Action (TRA).

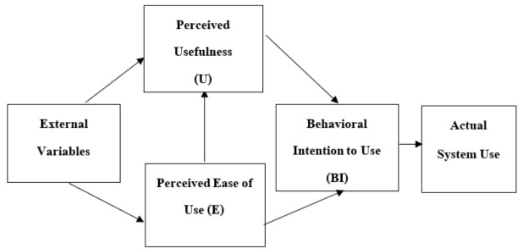

Fig. 2. Theory of Reasoned Action (TRA).

It is important to first raise awareness about the benefits of technology. This will help public health volunteers have clear expectations about the benefits they will receive and then decide whether to use them continuously or not. By fostering a positive attitude toward technology, it has been found that the Technology Acceptance Model (TAM) developed by (Davis,1985) is the most widely used model to measure customer acceptance of technology. Furthermore, the Technology Acceptance Model (TAM) categorizes factors related to technology acceptance into two main issues: perceived benefits and perceived ease of use. These two factors influence the development of attitudes toward technology, which can lead to feelings of satisfaction stemming from a positive attitude. This information was provided in the previously mentioned literature review.

B. Theory of Aging

The World Health Organization defines the elderly as people aged 60 years or older [8]. It also includes individuals who have retired from work due to economic conditions or those who society recognizes as elderly due to societal and cultural norms. do not alter them.

At the 1982 World Assembly, the United Nations defined "elderly people" as individuals aged 60 years and over, including both males and females.

According to Bunlu Siripanich (2001) "elderly" refers to a person who is born, grows up to be a child, then an adult, and finally an elderly person. We also refer to them as "old people" or "seniors." However, closer inspection reveals that we can separate the terms used to refer to the elderly as follows:

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1. Physical characteristics indicate old age, leading to terms like "old person," "elderly person," or "old man."

2. The term 'elderly' refers to people who are older according to the calendar, typically in Western and European cultures, as well as in America, where individuals aged 65 years and over are often considered elderly." However, in the Asian region, individuals 60 years and above are considered elderly, despite the international consensus that they should be considered senior citizens.

3. According to social status, when addressing senior members of an organization, the leader is considered an authority figure within the organization, regardless of their age.

In conclusion, the term "elderly" refers to individuals who are 60 years of age and older, including both males and females. This group can be further divided into two categories: the "early elderly," which includes those aged 60 - 69, and the "late elderly," which encompasses individuals aged 70 and older. These age groups include both men and women.

The segmentation of the elderly according to the elderly health assessment form. The following three categories apply to elderly people:

1) Social bound. They are socially active and self-sufficient elderly individuals. They are capable of helping others while also contributing to society and the community. They can walk upstairs unassisted, leave the house, and sleep independently. They can walk alone on flat surfaces, eat without assistance, and use the toilet successfully.

2) Home bound. Elderly people may need assistance walking on smooth surfaces and performing daily tasks like eating and using the bathroom.

3) Bed bound. Elderly individuals who are unwell and unable to care for themselves, as well as those with disabilities, may experience difficulty moving around while sitting or lying down. They may also struggle with swallowing food and may require assistance from a caregiver. Additionally, they may need to wear diapers and have them changed regularly.

C. The Concepts of Public Health Volunteer.

Public Health Volunteers are individuals selected from villagers, each responsible for at least 10 households in their neighborhood. The Ministry of Public Health specifies its training curriculum. In addition, they play a crucial role as change agents, leading the way in altering health behaviors and serving as public health communicators. They provide guidance on disseminating knowledge and coordinating public health development activities. Moreover, the village/community provides public health services to its households. On average, 1 person is responsible for 10 – 15 households.

Duties and responsibilities of public health volunteers include the following [9].

1) Promote healthy behaviors by serving as a role model and encouraging the community to engage in activities that promote health, control disease, and prevent illness. Take the lead in organizing community initiatives and inviting neighbors to participate in activities aimed at enhancing health and overall quality of life.

2) Take action to protect the environment by advocating for global warming reduction. Additionally, strive to prevent and control infectious diseases, and assume a leading role in addressing health issues within the community.

3) Engage in management, planning, problem-solving, and community development.

4) Public health media serves as a bridge between officials and the village's residents. The community can become a powerful health communicator.

5) Coordinate with individuals, organizations, and health partner groups.

6) Ensure the provision of health insurance and public health benefits for the villagers by taking a leadership role and collaborating with community leaders.

7) Deliver public health services to the community, including basic medical care, and provide follow-up and support for patients referred from the healthcare facility.

8) Work at the primary health care center or a designated location in the village.

D. Related Research and Article.

Nopparit Srithiang and Pichitchai Putklang (2019) have developed application systems and equipment to track elderly behavior by sensing activity from accelerometers and gyroscopes and extracting features (feature extraction) [10]. The application uses machine learning to classify activities during the pre-processing phase of data preparation, before algorithm processing. The application then displays the activity results, alerting users when an elderly person stumbles. This work will allow users to assist the elderly as soon as possible and facilitate scheduling doctor appointments for them.

Suparat Kaewserm (2020) conducted a study on developing smartphone applications for elderly care. The application's goal was to expand medical consultation options for the elderly. For instance, during the "COVID-19" pandemic, patients or their caregivers could use the application to seek medical advice and request emergency medical assistance. The application could also provide navigation guidance for the medical vehicle to reach the patient's location based on specified coordinates. The researcher evaluated user satisfaction with the application among the sample group. The findings revealed that the respondents expressed overall satisfaction with the smartphone application for elderly care, with an average satisfaction level of $\bar{x}$ 4.46 and a standard deviation of S.D. = 0.57 [11].

Pawaphan Khamthab and Teeraphong Wiriyanon (2022) researched the design of multiple user interfaces based on digital technology in an aging society. The objective is to design a user interface for the elderly. We collected data using a usability assessment form. The research revealed that the user interface design for the elderly is of high quality. It includes a graphical, touch, and audio user interface [12].

Thosaporn Sangkangwan (2022) conducted a research study focused on designing and developing applications to care for the welfare of the elderly. The goal was to test the application's efficiency in caring for the elderly's welfare. Three experts conducted testing to assess the quality of the application's design and performance, which also included an evaluation of user satisfaction. The sample population consisted of 50 elderly people living in Bangkok. The results showed that the sample group was satisfied with the application's design, with an average score of 4.04, indicating a satisfactory level. In terms of efficiency, the average score was 4.0, also indicating a satisfactory level. This illustrates the application's feasibility [13].

The researcher found that application development is most effective when elderly individuals and public health volunteers perceive useful applications as simple to use, based on a review of related literature. This awareness leads to a positive attitude among the elderly toward using the application. Over time, this positive attitude fosters the formation of behaviors and intentions to use the application, ultimately leading to its actual use. This finding aligns with the Technology Acceptance Model (TAM) mentioned earlier.

III. Research Methodology

A. Data Collection

In this research, the researcher collected data through in-depth interviews to gain detailed information and address the research objectives. The interviewees included 3 high-level and middle-level executives from Public Health Center 51, Chatuchak District, Bangkok, and 15 public health volunteers. We also collected data using a questionnaire to evaluate the effectiveness and satisfaction of the application. A total of 48 respondents participated, including 30 elderly individuals, 15 public health volunteers, and 3 information technology experts.

This research involves qualitative analysis using a method of data categorization known as classification. The data is obtained from in-depth interviews. Quantitative analysis is also conducted using statistical methods to evaluate the efficiency and satisfaction of using the application. The statistical methods employed involve determining the frequency and percentage of the obtained data, as well as calculating the means and standard deviation.

The researcher applies the principles of the System Development Life Cycle (SDLC) theory to the development of the system, which consists of five steps as follows [14].

1) Planning. The researcher actively engaged with the community to gather vital information, analyze pertinent issues, and identify specific needs. We conducted in-depth interviews with the sample group, selected key community problems for our research focus, examined the underlying causes of these issues, and explored relevant principles, theories, associated research and development programs, and user interface design.

2) Analyze and design. The researcher uses the principles of object-oriented programming and a use case diagram to illustrate the system's overall functionality and the user's interaction with it. Moreover, the researcher categorizes the users into three groups : public health volunteers, the elderly, and the system administrators. The users can apply for the membership and access information about the elderly. Figure 3 provides details on use case diagram.

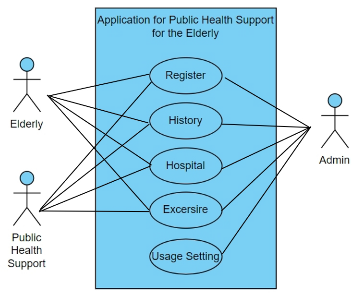

Fig. 3. Use Case Diagram.

However, we strictly protect user personal information through data encryption, user authentication, and a secure data management system to prevent unauthorized access. Additionally, the researcher developed a class diagram that illustrates the system's classes and their relationships. This diagram includes four groups : administrators, members, the elderly people, public health volunteers, hospitals, and exercise programs, as shown in Figure 4.

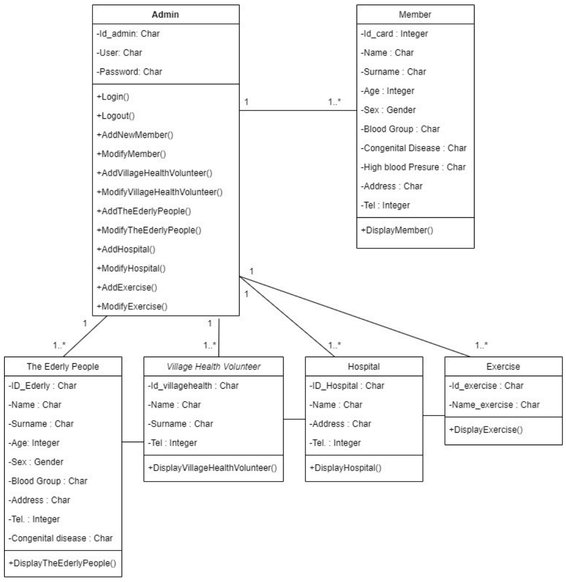

Fig. 4. Class Diagram.

3) Development. The researcher creates development applications using programming languages like ASP.NET and Java, as well as databases like SQL Server. Additionally, we simulate a server using PHP Apache.

4) Testing. The researcher tested the interoperability of functions and found that all applications worked correctly, and the data was reliable. The tester was an information technology expert, along with public health volunteers and the elderly. After completing the application, the researcher prepared an official letter from Chandrakasem Rajabhat University to Google Play Thailand Company. This letter attests to the application's inclusion in the research and guarantees the confidentiality of users' personal information. The researcher then submitted the developed application and database to the Google Play system.

5) Maintenance. For Enhancing Performance (Perfective Maintenance). This application aims to improve performance and enhance user experience, making it easier and more convenient to use 24 hours a day. In addition, the researcher has prepared a user manual and training on the use of the application for public health volunteers and the elderly. This allows users to use this application with more confidence and safety.

IV. Results

We have developed this application for the Android operating system. It can be downloaded from the Google Play using the PHS Application.

Fig. 5. PHS Application can be downloaded from Google Play.

A. *Result for this application*

The application's menus include support public health assistance for elderly individuals. The application has 4 main menu functions.

1) History menu: Stores personal information about the elderly, such as congenital diseases, treatment rights, and treatment history as shown in Figure 6 – 7.

Fig. 6. Application main menu.

Fig. 7. History menu.

2) Hospital menu: Displays information on the names of public health service centers and hospitals in Chatuchak, Bangkok as shown in Figure 8.

Fig. 8. Hospital menu.

3) Exercise menu: Display exercise video clips for the elderly.

4) Usage settings menu: Show the application's usage settings.

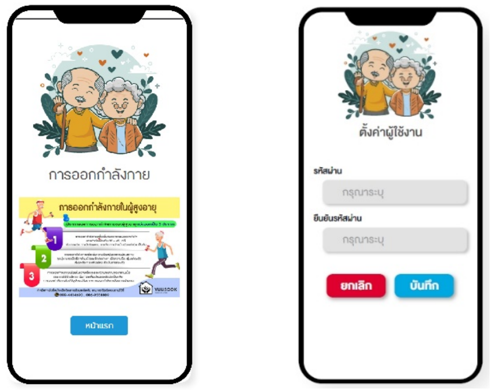

Fig. 9. Exercise menu and Usage setings menu.

B. *Evaluation results for efficiency and satisfaction*

The Development of the Application for Public Health Support for the Elderly population enhances its effectiveness and satisfaction by enabling users to test its various functions and experience its operation. Then let users evaluate their satisfaction with the application system. We utilized the feedback from the evaluation to establish guidelines for enhancing and refining the system. The research questionnaire uses the interval scale type and employs the Likert scale [15] to rate satisfaction levels. The assessment involved a total of 48 people. The evaluation revealed 30 elderly individuals, 15 public health volunteers, and 3 information technology experts. Different groups categorized the findings, with 21 males representing 43.75 percent and 27 females representing 56.25 percent. The user groups consist of the following: elderly (30 people, 62.50%), public health volunteers (15 people, 31.25%), and information technology experts (3 people, 6.25%).

TABLE I. RESULTS OF EVALUATION OF THE EFFICIENCY AND SATISFACTION OF THE APPLICATION.

Item	$\bar{x}$	(S.D)	Satisfaction
1. Application performance	4.67	0.48	The most
2. Application design	4.47	0.57	high
3. Proper application functionality	4.40	0.56	high
4. User interface friendly	4.30	0.47	high
5. Clear content explanations	4.53	0.57	The most
6. The application's aesthetic appeal	4.60	0.62	The most
7. Appropriate font size	4.63	0.49	The most
8. Proper use of text colors	4.43	0.63	high
9. The application makes appropriate use of colors.	4.57	0.50	The most
10. Pictures allow for clear species identification.	4.60	0.56	The most
11. Suitable placement of elements	4.77	0.43	The most
12. The application is helpful for the user.	4.73	0.52	The most
13. The application fulfills the user's needs.	4.73	0.45	The most
Total	4.57	0.54	The most

The results of the evaluation of the efficiency and satisfaction with the application were as follows: the mean was 4.57, the standard deviation was 0.54, which is statistically considered to be at the highest level of satisfaction.

V. Discussion

Researchers have developed an application named PHS Application for the Android operating system, according to their findings. You can download this app from the Google Play to support public health assistance for the elderly population. Elderly people and public health volunteers use this application. The application features a history menu, hospital menu, exercise menu, and usage settings menu. It enables the elderly to record their history and closely monitor their illness status. There are 3 statuses: normal, watch, and sick. In an emergency, the elderly can request help by pressing the emergency button in the application. Pressing the button sends a notification to all public health volunteers, enabling them to visit the elderly's homes and provide basic assistance before transporting them to a hospital in case of illness or accident.

In addition, the researcher assessed the effectiveness of the application with the three information technology experts. The researcher measured the satisfaction with the application among 30 elderly individuals and 15 public health volunteers, resulting in 48 participants who could explore its various functions. The evaluation indicated that users expressed the highest level of satisfaction with the application, largely due to its appealing presentation and well-designed user interface, which featured appropriate font sizes and content formatting. Additionally, the application included video clips and user manuals presented in infographic format, specifically developed to cater to the needs of the elderly, particularly those with limited technological skills.

Additionally, public health volunteers and elderly users of the application have recognized its benefits and developed a positive attitude towards its use. This aligns with the Technology Acceptance Model (TAM), as post-implementation assessments indicate a favorable response. Notably, public health volunteers have demonstrated quick response times, assisting an elderly community member who experienced a fall and another individual with high blood pressure who needed to be taken to the hospital.

This research concludes that the developed application is highly effective in supporting and promoting the use of technology for the elderly, particularly in healthcare, and in providing effective access to necessary information. Waraporn Ariyasat and Phanuwat Atthanasak focused their study on developing training packages of Google applications for Learning [16]. To assess technology adoption, the study used a questionnaire based on the Technology Acceptance Model (TAM) [5]. The research results, based on the training package trial, showed that both students and the community widely accepted and utilized Google applications at a high level.

Additionally, Chulawalee Moneelert has completed the development of application for Bed-Bound older adults care promotion via augmented reality technology. The research results indicate that users are highly satisfied with the application and fully accept its use. You can use the application to promote elderly care [17].

VI. Conclussion

The researchers to utilize the developed application, and the researchers publicize to other communities that are interested in this project. Additionally, we have expanded the network to include both public and private hospitals in nearby communities and aimed to enhance its utilization throughout Thailand. Future research can further develop applications in other elderly-related contexts, such as assessing the ability to perform daily activities in the elderly group of dependent people. Monitoring falls in the elderly We should develop applications that use medicine, food, or nutrition to promote health for the elderly in other areas. We need to develop applications for the iOS operating system to facilitate a wider range of tasks.

Acknowledgment

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